DO YOU KNOW?

Synthetic analogs of purines, pyrimidines, nucleosides and nucleotides or sugar moiety have numerous application in clinical medicine. Oncologists employ 5- fluoro - or 5 iodouracil, and 8-azaguanine for treatment of cancer. The purine analog allopurinol is used in treatment of hyperuricemia and gout. Azathioprine is employed during organ transplantation to suppress immunologic rejection.

7.5.2 *Salient features of the structure of DNA*

The B form DNA, also known as the Watson- Crick DNA is the most stable and prevalent form of DNA. The important structural features of B- DNA are :

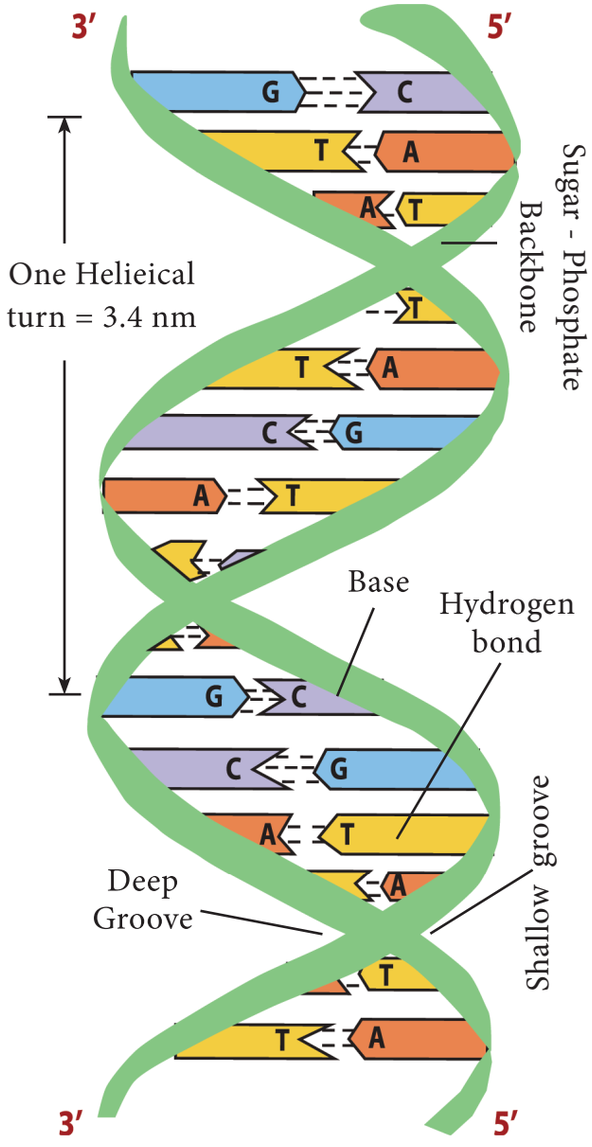
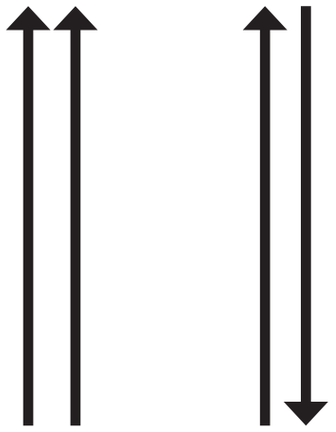


Figure. 7.6 Sructure of DNA

1. Tvhere are two polynucleotide chains in the DNA spirally twisted around each other to form a right handed double helix.



Parallel **Anti-parallel**

2. The sugar-phosphate backbones remain on the outside, while the core of the helix contains the purine and pyrimidine bases.

3. The diameter of DNA is 2 nm or 20 A°. The length of a complete turn of helix is 3.4 nm or 34 A° i.e. there are ~10.5bp per turn.

4. The DNA helix has a shallow groove called minor groove (-1.2nm) and a deep groove called major groove (-2.2nm). Proteins interact with DNA through the minor and major grooves without disrupting the DNA strands.

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